

Fourier Series Representation of Periodic Signals

1 The Response of LTI Systems to Complex Exponentials

Definition 1.1 (Eigenfunctions & Eigenvalues).

A signal for which the system output is a constant times the input is referred to as an *eigenfunction* of the system, and the amplitude factor is referred to as the system's *eigenvalue*.

It is easy to show that *complex exponentials* are eigenfunctions of LTI systems.

$$e^{st} \xrightarrow{\text{a CT LTI system}} H(s)e^{st} \qquad z^n \xrightarrow{\text{a DT LTI system}} H(z)z^n$$

where $H(s) := \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$ or $H(z) := \sum_{k=-\infty}^{\infty} h[k]z^{-k}$ is called the *transfer* or *system function* of the system.

Proof.

$$h(t) * (e^{st}) = \int_{-\infty}^{\infty} h(\tau)e^{s(\tau-t)}d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$$

where $H(s)$ is the eigenvalue. □

Therefore, we have

$$x(t) = \sum_i A_i e^{s_i t} \xrightarrow{\text{an LTI system}} y(t) = \sum_i H(s_i) A_i e^{s_i t}$$

In the following sections, we only focus on the purely imaginary values of s , i.e., $s = j\omega$.

2 FS Repr of CT Periodic Signals

All CT complex exponential signals that are periodic with period T is given by

$$\phi_k(t) := e^{jk\omega_0 t} = e^{jk\frac{2\pi}{T}t} \quad k \in \mathbb{Z}$$

where $\omega_0 = 2\pi/T$.

Many periodic signals can be represented as a linear combination of harmonically related complex exponentials:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

We choose harmonically related complex exponentials due to their orthogonality:

$$\langle e^{jm\omega_0 t}, e^{jn\omega_0 t} \rangle = \int_T e^{j(m-n)\omega_0 t} dt = \begin{cases} T & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

where $m, n \in \mathbb{Z}$ and $T = \frac{2\pi}{\omega_0} > 0$.

Definition 2.1 (The FS of a CT Periodic Signal).

If a signal $x(t)$ with a period T has a Fourier series representation, then

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t} \quad \text{(The Synthesis Equation)}$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk\frac{2\pi}{T}t} dt \quad \text{(The Analysis Equation)}$$

where the set $\{a_k\}$ is called the *Fourier series coefficients* or the *spectral coefficients* of $x(t)$.

Remark. a_0 is simply the average value over one period, called the *dc* or *constant component* of $x(t)$.

$$a_0 = \frac{1}{T} \int_T x(t) dt$$

The coefficients can be seen as projections of the signal onto the eigenfunctions of the system:

$$a_k = \frac{\langle x(t), e^{jk\omega_0 t} \rangle}{\langle e^{jk\omega_0 t}, e^{jk\omega_0 t} \rangle} = \frac{\int_T x(t) e^{-jk\omega_0 t} dt}{\int_T e^{jk\omega_0 t} e^{-jk\omega_0 t} dt} = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

3 The FS of a DT Periodic Signal

All DT complex exponential signals that are periodic with period N is given by

$$\phi_k[n] := e^{jk\omega_0 n} = e^{jk\frac{2\pi}{N}n} \quad k \in \mathbb{Z}$$

where $\omega_0 = 2\pi/N$.

However, there are only N distinct signals given by this, because

$$\phi_{k+N}[n] = e^{jk\frac{2\pi}{N}n + jk2\pi} = e^{jk\frac{2\pi}{N}n} = \phi_k[n] \quad \forall k \in \mathbb{Z}$$

For a single ϕ_k , the following property holds:

$$\sum_{n=\langle N \rangle} \phi_k[n] = \sum_{n=\langle N \rangle} e^{jk\frac{2\pi}{N}n} = \begin{cases} N & \text{for } k = 0, \pm N, \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}$$

Proof.

$$\sum_{n=\langle N \rangle} e^{jk\frac{2\pi}{N}n} = \sum_{n=n_0}^{n_0+N-1} e^{jk\frac{2\pi}{N}n} = \begin{cases} \sum_{n=\langle N \rangle} 1 = N & \text{for } e^{jk\frac{2\pi}{N}} = 1 \\ \frac{e^{jk\frac{2\pi}{N}n_0} (1 - e^{jk\frac{2\pi}{N}N})}{1 - e^{jk\frac{2\pi}{N}}} = 0 & \text{otherwise} \end{cases}$$

□

Remark. “ $\sum_{n=\langle N \rangle}$ ” means the sum over any interval of N integers.

Orthogonality

$$\langle \phi_k[n], \phi_l[n] \rangle = \sum_{n=\langle N \rangle} \phi_k[n] \phi_l^*[n] = \sum_{n=\langle N \rangle} e^{j(k-l)\omega_0 n} = \begin{cases} N & \text{for } k = l \\ 0 & \text{for } k \neq l \end{cases}$$

where k, l are in the same period, i.e., $|k - l| < N$.

Definition 3.1 (The FS of a DT Periodic Signal).

If a signal $x[n]$ with a period N has a Fourier series representation, then

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} a_k e^{jk\frac{2\pi}{N}n} \quad \text{(The Synthesis Equation)}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\frac{2\pi}{N}n} \quad \text{(The Analysis Equation)}$$

where the set $\{a_k\}$ is called the *Fourier series coefficients* or the *spectral coefficients* of $x[n]$.

The coefficients are also periodic with period N , i.e.,

$$a_k = a_{k+N} \quad \forall k \in \mathbb{Z}$$

4 Properties of the Fourier Series

4.1

Suppose we have $x(t)$ and $y(t)$, or $x[n]$ and $y[n]$, of the **same period** T or N , and

$$\left\{ \begin{array}{l} x(t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_k \\ y(t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} b_k \end{array} \right. \quad \text{or} \quad \left\{ \begin{array}{l} x[n] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_k \\ y[n] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} b_k \end{array} \right.$$

Then we have the following results:

Linearity

$$A \cdot x(t) + B \cdot y(t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} A \cdot a_k + B \cdot b_k \qquad A \cdot x[n] + B \cdot y[n] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} A \cdot a_k + B \cdot b_k$$

Time Shifting

$$x(t - t_0) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} e^{-jk\omega_0 t_0} a_k \qquad x[n - n_0] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} e^{-jk\omega_0 n_0} a_k$$

Frequency Shifting

$$e^{jM\omega_0 t} x(t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_{k-M} \qquad e^{jM\omega_0 n} x[n] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_{k-M}$$

Time Reversal

$$x(-t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_{-k} \qquad x[-n] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_{-k}$$

Time Scaling

$$x(\alpha t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_k \left(\begin{array}{l} \text{with fundamental} \\ \text{frequency } \alpha\omega_0 \end{array} \right) \qquad \left\{ \begin{array}{ll} x[\frac{n}{m}] & \text{for } \frac{n}{m} \in \mathbb{Z} \\ 0 & \text{otherwise} \end{array} \right. \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} \frac{1}{m} a_k \left(\begin{array}{l} \text{with period} \\ \text{of } mN \end{array} \right)$$

Conjugation

$$x^*(t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_k^* \qquad x^*[n] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_k^*$$

Periodic Convolution

$$\int_T x(\tau) y(t - \tau) d\tau \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} T a_k b_k \qquad \sum_{r=\langle N \rangle} x[r] y[n - r] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} N a_k b_k$$

Multiplication

$$\left(\begin{array}{l} \text{of same} \\ \text{period} \end{array} \right) x(t) y(t) \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} a_k * b_k \qquad x[n] y[n] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} \sum_{r=\langle N \rangle} a_r b_{k-r}$$

Differentiation & Difference

$$\frac{dx(t)}{dt} \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} (jk\omega_0) a_k \qquad x[n] - x[n - 1] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} (1 - e^{-j\omega_0}) a_k$$

Integration & Running Sum

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} \frac{1}{jk\omega_0} a_k \qquad \sum_{k=-\infty}^n x[k] \xleftrightarrow[\mathcal{FS}^{-1}]{\mathcal{FS}} \frac{1}{1 - e^{-jk\omega_0}} a_k$$

Parseval's Relation for Periodic Signals

$$\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2 \qquad \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$

4.2 Symmetry

Both CTFS and DTFS have exactly the same symmetry properties as the CTFT, including conjugate symmetry, parity, and even-odd decomposition.

5 Transfer Functions

For a given system, H is called the **transfer function**, **system function**, or **frequency response**.

5.1 CT Periodic Signals

Definition 5.1 (Transfer Function of a CT System).

Suppose $h(t)$ is the *impulse response* of the system, then the transfer function H is given by

$$H(j\omega) := \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt$$

Suppose $x(t)$ is a periodic signal and a_k is the FS representation of $x(t)$.

$$y(t) = x(t) * h(t) \xrightarrow{\mathcal{FS}} a_k \cdot H(jk\omega_0)$$

Proof.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \xrightarrow{\mathcal{FS}} y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

□

Definition 5.2 (Transfer Function of a DT System).

Suppose $h[n]$ is the *impulse response* of the system, then the transfer function H is given by

$$H(e^{j\omega}) := \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

Suppose $x[n]$ is a periodic signal and a_k is the FS representation of $x[n]$. (a_k is periodic.)

$$y[n] = x[n] * h[n] \xrightarrow{\mathcal{FS}} a_k \cdot H(e^{jk\omega_0})$$